

LIQUID-LIQUID EXTRACTION

Group # 1

Diego Martinez

Allison Nordvall

Peter Starakiewicz

Emily Sobie

James Fell

William Rawson

Chemical Engineering Lab II

3/3/2020

Introduction

- **Objective**
 - The emphasis of the lab is to study the characteristics of liquid-liquid extraction using a three-stage mixer-settler system.
 - Measure the concentration of acetic acid as a function of stage number and agitator speed for the two phases.
 - Use a McCabe-Thiele diagram to obtain the number of theoretical stages
 - Determine the Murphree efficiencies for the stages, and to calculate the overall efficiency

Theory

- The most common equation used in the experiment is

$$M_1V_1 = M_2V_2$$

- Using the concentration of acetic acid in the phases, the following formula can be used to plot the equilibrium curve

$$y_{n+1} = \left(\frac{E^{n+1} - 1}{E - 1} \right) y_1$$

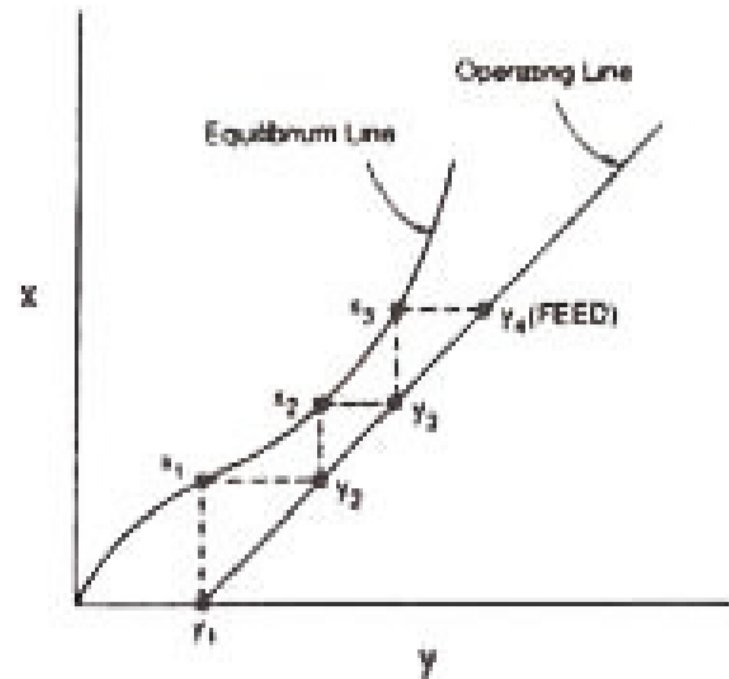


Figure 1: Multi-stage Extraction curve

Theory

- Murphree efficiency equation for the stages
 - $E_{ME} = \frac{Y_{S+1} - Y_S}{Y_{S+1}^* - Y_S}$
- Murphree efficiency equation for the raffinate phase
 - $E_{MR} = \frac{X_S - X_{S+1}}{X_S - X_{S+1}^*}$
- Overall efficiency of the solvent extraction
 - $E_0 = \frac{\eta_T}{\eta_{act}}$

Apparatus



1	Tanks	The white oil and acetic acid mixture will be in the feed tank at the beginning of the experiment.
2	Agitator	There is an agitator in each of the three stages, to ensure that the stage is mixed well.
3	Mixer-settler Stages	The bins are where the mixing of the liquid-liquid extraction occurs. There are three
4	Switch	This switch is used to power the three agitators.
5	Switch (pump)	This switch is used to power the pump.
6	Recycle Valve	This recycle valve is used to recycle the water to create a closed circuit.
7	Drain	There are a total of 2 drains that are present on each bin, for a total of 6 drains.
8	Water Valve	This is used to allow the water to enter the system.
9	Water Rotameter	This is used in order to adjust flow rate of water that is being put into the system
10	White Oil Rotameter	This is used in order to adjust the flow rate of white oil
11	Pump	This is used in order to pump the white oil into the three different stages.
12	Agitator Dials	These dials are used to vary the speed of the agitator. There is one on each agitator
13	White Oil Flow Valve	This is used to allow the white oil mixture to enter the system.
14	Product Return Pipe	This pipe returns the product to the feed tank. This will only be operated by the instructor.

Material & Supplies

1	TG 55 White Mineral Oil	25 gallons of white mineral oil will be mixed with acetic acid.
2	Glacial Acetic Acid	0.5 wt% will be extracted from the white mineral oil.
3	Water	Water is the solvent used to separate acetic acid from the white mineral oil.
4	0.05 M NaOH	Will be used to determine the concentration of acetic acid in the white mineral oil and water.
5	Phenolphthalein	Phenolphthalein will be the indicator used in the titration process.
6	50 mL Burette with Clamp	Will be used to hold the 0.05 M NaOH solution to perform titrations.
7	Serologic Pipet	Will be used to transfer the sample of white mineral oil and water for future titrations.. The size of the pipet is 25 mL in 2/10 intervals.
8	Beaker/Erlenmeyer Flask	Used to store samples of white mineral oil and water before and during titration.
9	Rubber Bulb	Used to suck the sample of white mineral oil and water into the serologic pipet.
10	Stir Bar	This is used to stir the sample mixture.
11	Chemical Resistant Gloves	Because glacial acetic acid and 0.05 M NaOH will be used, the gloves are needed for protection.

Procedure

- Ensure feed tank is filled to around 25 gallons, otherwise ask instructor for instance
- Perform a titration on a sample of the oil
 - Calculate and add the amount of Acetic Acid needed to reach 0.5 wt%
- Mix the Acetic Acid and the White oil in the feed tank by using the centrifugal pump in the closed circuit.
 - Take a sample of the mixture to determine the composition of acetic acid within the white oil.
- Open the water flow valves until the rotameter shows a flowrate of 0.7 GPM.
- Adjust the speed of the three agitators to 400 rpm each.
- Once the water circuit has been established, the White Oil feed will be fed into the system at around 0.4 GPM
- Samples of White oil and water phases will be withdrawn from each stage and the composition of Acetic Acid within them will then be determined by performing a titration with 0.05 NaOH
- The experiment will then be performed three times with agitator speeds of 600 RPM and 800 RPM

Experimental Challenges

- Mixing of the oil and water during titration. Must use a flask, stir bar, and a high stir rate when titrating.
- Constantly check to make sure your water flow rate is at 0.70 GPM as it fluctuates if other people are also using the water supply.
- Check the oil flow rate to make sure it is 0.4 GPM as it fluctuates as well.
- When using the tachometer, you must stabilize the device and point directly at the tape on the spinning impeller.

Safety

- **Potential Electrical Hazards**

- The electric motors, wires, and electrical box can pose shock hazards if they contact water.

- **Potential Mechanical hazards**

- The rotating impellers can create pinch points if body parts contact them while they are rotating. The mineral oil can make the floor slippery increasing possibility of a fall hazard.

- **Potential Pressure Hazards**

- None

- **Potential Temperature Hazards**

- The electrical motors can get hot, take caution when touching the motors when they are on or have recently turned off.
- **Hazardous/Toxic Chemicals**
- Sodium hydroxide, glacial acetic acid and phenolphthalein will be used in this experiment.
- NaOH and glacial acetic acid could cause severe burns of eyes, skin, and mucous membranes.
- Phenolphthalein is carcinogenic and an eye irritant.